

SR UNIVERSITY

AI ASSISTED CODING

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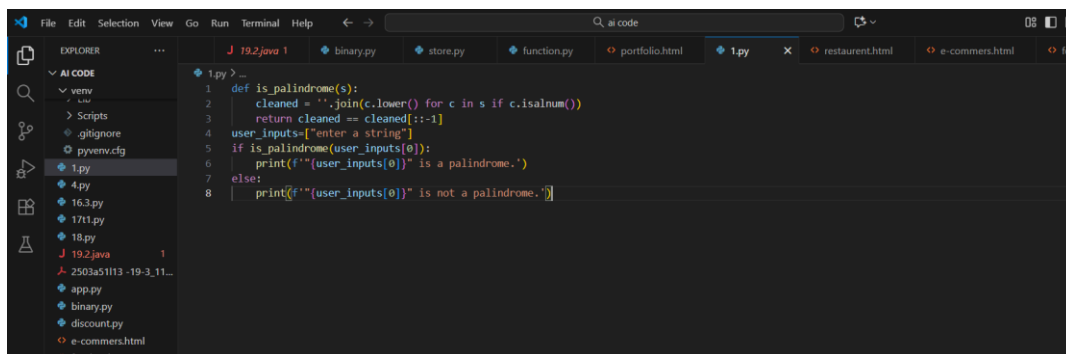
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TASK #1:

Prompt Used:

Write a comment: # Function to check if a string is a valid palindrome (ignoring spaces and case) and allow Copilot to complete it.

Code Generated:



```
1.py > ...
1 def is_palindrome(s):
2     cleaned = ''.join(c.lower() for c in s if c.isalnum())
3     return cleaned == cleaned[::-1]
4 user_inputs=["enter a string"]
5 if is_palindrome(user_inputs[0]):
6     print(f'{user_inputs[0]} is a palindrome.')
7 else:
8     print(f'{user_inputs[0]} is not a palindrome.')
```

Output:

```
PS C:\ankitha\ai test > & 'c:\Program Files\Python313\python.exe' 'c:\Users\ADHARSH\.vscode\extensions\ms-python-bugpy\launcher' '57046' '--' 'c:\ankitha\ai test 3\1-t1.py'
"enter a string" is not a palindrome.
```

Observations:

- It cleans the input by removing all non-alphanumeric characters and converting everything to lowercase, ensuring accurate results regardless of punctuation, spacing, or case.
- Checks for palindrome by comparing the cleaned string to its reverse (cleaned[::-1]). If both match, the string is a palindrome.

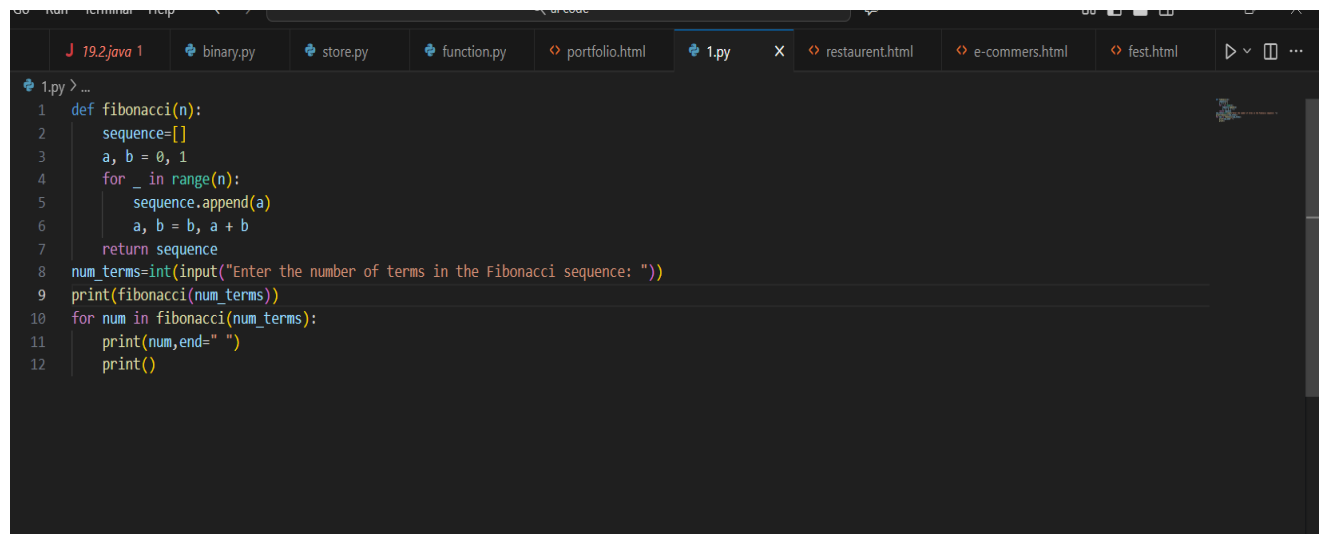
- Works for sentences and phrases (not just single words), as spaces, punctuation, and cases are ignored in the check.

TASK #2:

Prompt Used:

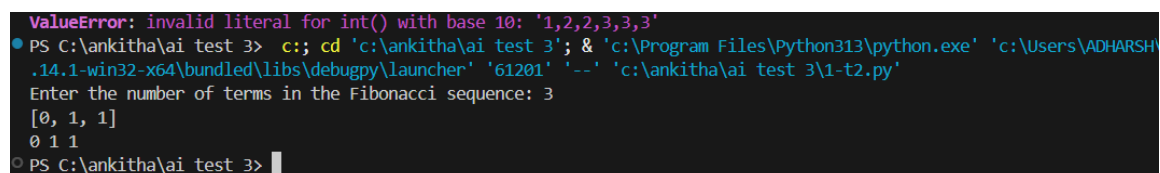
Generate a Python function that returns the Fibonacci sequence up to n terms. Prompt with only a function header and docstring.

Code Generated:



```
1 def fibonacci(n):
2     sequence=[]
3     a, b = 0, 1
4     for _ in range(n):
5         sequence.append(a)
6         a, b = b, a + b
7     return sequence
8 num_terms=int(input("Enter the number of terms in the Fibonacci sequence: "))
9 print(fibonacci(num_terms))
10 for num in fibonacci(num_terms):
11     print(num,end=" ")
12     print()
```

OUTPUT:



```
ValueError: invalid literal for int() with base 10: '1,2,2,3,3,3'
PS C:\ankitha\ai test 3> c::; cd 'c:\ankitha\ai test 3'; & 'c:\Program Files\Python313\python.exe' 'c:\Users\ADHARSHV
.14.1-win32-x64\bundled\libs\debugpy\launcher' '61201' '--' 'c:\ankitha\ai test 3\1-t2.py'
Enter the number of terms in the Fibonacci sequence: 3
[0, 1, 1]
0 1 1
PS C:\ankitha\ai test 3> |
```

Observations:

- The code generates the Fibonacci sequence up to a user-specified number of terms using a simple iterative approach.

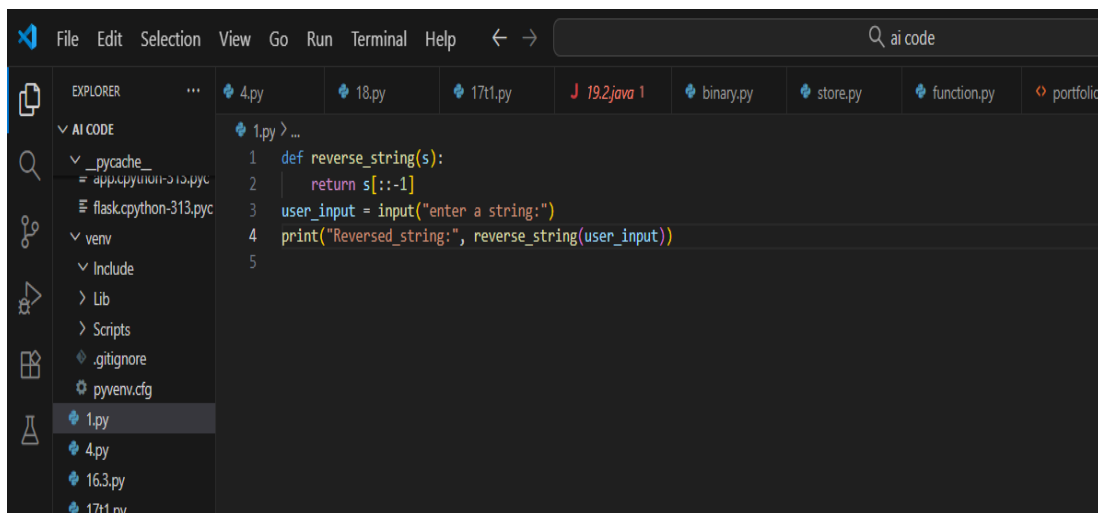
- The sequence is generated by initializing the first two terms ($a = 0$, $b = 1$) and iteratively updating them with $a, b = b, a + b$, ensuring each new term is the sum of its two immediate predecessors.
- Each generated term is stored in a list, which is then returned and printed.

TASK #3:

Prompt Used:

Write a comment like # Function to reverse a string and use Copilot to generate the function.

Code Generated:



```

1.py > ...
1 def reverse_string(s):
2     return s[::-1]
3 user_input = input("enter a string:")
4 print("Reversed string:", reverse_string(user_input))
5

```

Output:

```

PS C:\ankitha\ai test 3> c;; cd 'c:\ankitha\ai test 3'; & 'c:\Program Files\Python313\python.exe' 'c:\Users\ADHARSH\.vscode\extensions\ms-python.python-2023.14.1-win32-x64\bundled\libs\debugpy\launcher' '59935' '--' 'c:\ankitha\ai test 3\1-t3.py'
Enter a string: ankitha
Reversed string: ahtikna

```

Observations:

- The function uses Python's string slicing syntax `s[::-1]` to reverse the string.
- The slice step of `-1` means characters are taken from end to start, effectively reversing the string.

- User input is taken and passed to the function, with the reversed result printed.

TASK #4:

Prompt Used:

Generate a program that simulates a basic calculator (add, subtract, multiply, divide). Write the comment: # Simple calculator with 4 operations and let AI complete it.

CODE GENERATED:

```
1 # Simple calculator with 4 operations
2
3 def calculator(a, b, operation):
4     print('select operation:')
5     print('1. Addition')
6     print('2. Subtraction')
7     print('3. Multiplication')
8     print('4. Division')
9     choice = input('Enter choice (1/2/3/4): ')
10    num1=float(input("Enter first number: "))
11    num2=float(input("Enter second number: "))
12    if choice == '1':
13        print("result:", num1 + num2)
14    elif choice == '2':
15        print("result:", num1 - num2)
16    elif choice == '3':
17        print("result:", num1 * num2)
18    elif choice == '4':
19        if num2 != 0:
20            print("result:", num1 / num2)
21        else:
22            print("Error: Division by zero")
23    else:
24        print("Invalid input")
```

OUTPUT:

```
PS C:\ankitha\ai test 3> c:: cd 'c:\ankitha\ai test 3'; & 'c:\Program Files\Python313\python.exe' 'c:\Users\ADHARSH\.vscode\extensions\ms-python
.14.1-win32-x64\bundled\libs\debugpy\launcher' '52479' '--' 'c:\ankitha\ai test 3\t4.py'
Select operation.
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter choice (1/2/3/4): Addition
Enter first number: 6
Enter second number: 9
Invalid input
```

Observations:

- The function uses Python's string slicing syntax `s[::-1]` to reverse the string.
- The slice step of `-1` means characters are taken from end to start, effectively reversing the string.
- User input is taken and passed to the function, with the reversed result printed.